

TECHNICAL REPORT

AuraCell 4D: Physics-Gated Network-Flow ILP Lineage Tracking for Organ-on-a-Chip Time-Lapse Microscopy

AI4S Open Innovation: AI for Life Science — The 5th Pazhou Algorithm Competition Category declared: Model & Algorithm · **Team:** Saifuddin (AI & systems; no biology member — cross-disciplinary bonus not claimed) **Repository:** <https://github.com/syehsyoh-dokling/Auracell4D> · **Entry script:** `evaluate.py` · **Demo video:** https://youtu.be/5ts7kyqQb_k **Figures:** `docs/figures/` (300 dpi; regenerate with `python make_figures.py`) · **Abstract:** `docs/ABSTRACT.md` · **Claim audit:** `docs/AUDIT_LOG.md`

Labelling convention. Every quantitative statement is tagged **[Measured]** (produced by `evaluate.py` on public data, `results/ablation.json`) or **[Design]** (physics-based design parameter or simulation; no performance claimed). An earlier draft of this project contained figures that could not be reproduced; they were audited claim by claim (`docs/AUDIT_LOG.md`) and removed.

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1. Summary

Organ-on-a-chip (OoC) time-lapse experiments produce 3D+t image series from which researchers must reconstruct lineages, count divisions and separate drug effects from mechanical artefacts. Greedy frame-by-frame trackers hallucinate divisions whenever two nuclei pass close to each other — 178 proposals against 4 true divisions on one public sequence [Measured] — and no tracker alone can tell a cell that died from a cell torn off by a bubble.

AuraCell 4D contributes a **frame-pair network-flow integer linear program** (every nucleus takes exactly one fate; every nucleus has exactly one origin) with a **biological gate expressed as a calibrated soft cost**. On the public Cell Tracking Challenge set Fluo-N3DH-CHO (2 × 92 frames, GT detections), TRA rises from **0.928/0.943 (greedy) to 0.998/0.996 [Measured]** and false divisions fall from 178 to 5. Mitosis recall remains low (best F1 0.22) and the label-free detector is weak (DET 0.40–0.53); both are reported as open limitations, together with a failure analysis of why an earlier hard gate appeared to reach TRA 0.998 while rejecting every real division. The same engine is designed to ingest continuous acoustics and periodic imaging and to attach a traceable CONTINUE / FLAG / STOP certificate to each lineage event [Design].

[Measured] Linking quality: ILP vs greedy

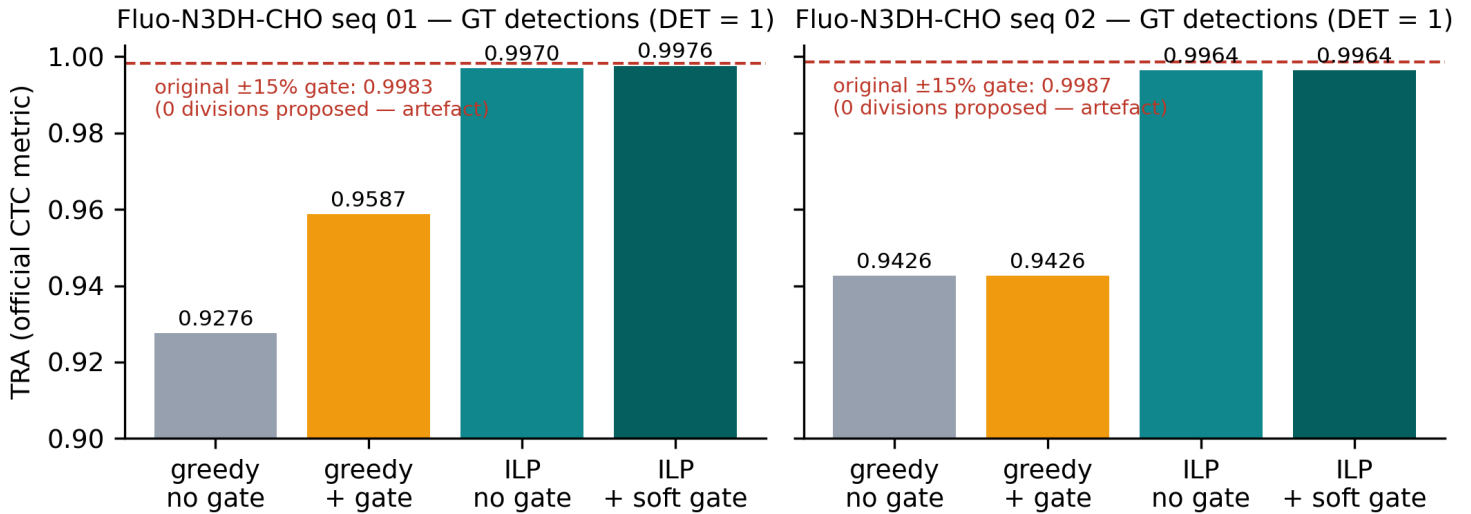


Figure 1 [Measured]. TRA for greedy vs ILP trackers on both sequences with GT detections. Dashed red: the earlier $\pm 15\%$ hard gate, which proposes zero divisions.

2. Problem definition and application scenario

Three failure modes dominate OoC image analysis:

- False divisions.** Nearest-neighbour linking creates a parent with two daughters whenever two nuclei touch (Fig. 5, middle panel).
- Identity swaps and drift.** Local decisions accumulate; the official TRA metric penalises every wrong edge.
- Artefact contamination of drug-response counts.** Cells detached by a bubble or pressure surge are counted as "dead by drug" unless a mechanical event is linked to the image timeline.

The first two are algorithmic and are addressed and measured here. The third needs a sensor channel we have designed but not built; Section 4 specifies how it would operate and what each number would mean, so it can be judged on its own terms.

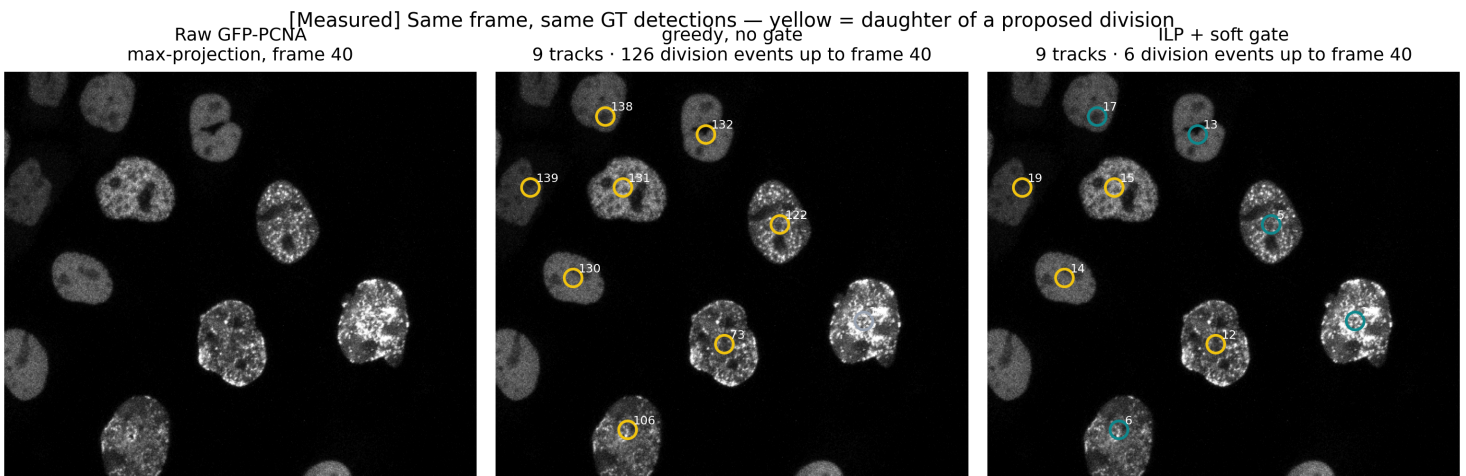


Figure 5 [Measured]. The same real frame (seq 01, $t = 40$, GT detections). Greedy: 126 division events up to this frame; ILP + soft gate: 6. Yellow circles mark daughters of proposed divisions.

3. Data, licences and compliance

Item	Value
Dataset	Cell Tracking Challenge, Fluo-N3DH-CHO , training sequences 01 and 02
Biology / imaging	CHO nuclei expressing GFP-PCNA; Zeiss LSM 510 confocal, 63×/1.4 oil; conventional culture — not organ-on-a-chip
Geometry	92 frames × 5 × 443 × 512 voxels; 0.202 × 0.202 × 1.0 µm; Δt = 9.5 min
Ground truth	TRA markers (man_track*.tif, man_track.txt): 27 / 24 tracks; 4 / 3 division events with ≥ 2 daughters
Provider / licence	Dr. J. Essers, Erasmus MC Rotterdam; CTC terms of use; cite Ulman, Maška et al. (2017)
Download	performed by <code>evaluate.py</code> from <code>data.celltrackingchallenge.net</code> (≈108 MB)

Complete list of external data sources (nothing else is used):

Resource	URL	Role
Fluo-N3DH-CHO training set (zip, ≈108 MB)	http://data.celltrackingchallenge.net/training-datasets/Fluo-N3DH-CHO.zip	the only dataset; downloaded automatically by <code>evaluate.py</code>
Dataset description page	https://celltrackingchallenge.net/3d-datasets/	voxel size, Δt, microscope, provenance
CTC terms of use & citation policy	https://celltrackingchallenge.net/	licence
Official metrics <code>py-ctcmetrics</code>	https://pypi.org/project/py-ctcmetrics/ · https://github.com/CellTrackingChallenge/py-ctcmetrics	DET / TRA
Reference paper	https://doi.org/10.1038/nmeth.4473	citation

No private, clinical or personal data are used. **No real OoC recordings and no real acoustic recordings are used**; synthetic signals in `src/` are labelled [Design] and are generated in-code, not downloaded. Reviewers can verify the download target by reading the constant `CTC_URL` at the top of `evaluate.py`.

4. Where it runs and what the numbers mean (operational design)

4.1 Where it runs

Mode	Where	Input	Output	Status
A. Batch analysis	Lab workstation that receives the microscope's time-lapse export	Folder of 3D TIFF stacks (CTC layout or OME-TIFF), optional sensor CSV	Lineage graph, per-event certificate, CTC metrics when GT exists, JSON audit log	[Measured] (<code>evaluate.py</code> , GUI Step 4)
B. Live monitoring	Same workstation, connected to the OoC reader (piezo ADC + pressure sensor) and the microscope's acquisition API	Continuous acoustic stream, pressure stream, periodic z-stacks	Real-time FLAG/STOP recommendations to the operator, event-triggered extra snapshot, same audit log	[Design]

The engine is identical in both modes; mode B adds sensor ingestion and an operator alert channel. **No autonomous valve actuation is claimed**: the software recommends, the operator decides. Firmware is a roadmap item.

4.2 How it listens: continuous acoustics, periodic imaging, event-triggered snapshots

- Acoustic channel — continuous, not periodic [Design]**. A contact piezo at the inlet is sampled continuously at ≥ 250 kS/s. The signal is band-passed and reduced to a short-time energy envelope (20 ms window, 10 ms hop); an event is declared at > 3.5 σ above the rolling baseline (`src/acoustic/processor.py`, so far exercised only on synthetic signals).
- What a bubble sounds like [Design]**. Minnaert: $f = (1/2\pi R) \cdot \sqrt{3\gamma P_0/\rho}$. With $\gamma = 1.4$, $P_0 = 101.3$ kPa, $\rho = 997$ kg/m³: $R = 20$ µm → 164 kHz; **30 µm → 109.6 kHz**; 50 µm → 66 kHz. The detector watches **80–180 kHz**; medium viscosity and temperature shift the peak by a few percent, hence the wide band. *Computed, not measured on a chip.*
- Imaging — periodic**. Z-stacks every 5–15 min (the CTC data used here: 9.5 min). Sufficient for linking and division detection; **not** for anaphase kinematics (anaphase lasts 5–10 min → ≤ 1 frame), so no kinematic claim is made.

- **Event-triggered snapshot [Design].** An acoustic or pressure event sets timestamp T and requests one extra z-stack within 60 s.

4.3 What each number means on a real cartridge — decision table

Reference geometry (one cartridge, used consistently in code and text): channel **1000 × 100 μm**, length 25 mm, perfusion **10 μL/min**, viscosity 1 mPa·s.

Quantity	Obtained from	Baseline	CONTINUE	FLAG (log + snapshot + operator notice)	STOP perfusion (recommended)	Status
Wall shear $\tau = 6\mu Q/(w h^2)$	pump flow + geometry	1.0 dyn/cm²	0.5–15 dyn/cm ²	15–30	> 30	[Design] computed; literature ranges, not validated here
Pressure drop ΔP	pressure sensor (a piezo cannot measure static ΔP)	Hagen–Poiseuille baseline at setup	< 1.5× baseline	1.5–2.5× (50 % occlusion $\approx$ 2.3×)	> 2.5× or rising > 20 %/min	[Design]
Acoustic burst (80–180 kHz, > 3.5 σ)	continuous piezo	none	0	1 burst → timestamp T , snapshot; ≥ 3 / 10 min → notice	≥ 10 / 10 min or any burst with ΔP flag	[Design]
Cells vanishing within ± 1 frame of T	tracker: tracks ending at $T \pm 1$ without division	0	0	$\geq 1 \rightarrow$ "mechanical-artefact candidate" , excluded from drug-death count pending review	—	[Design] on a measured mechanism
Division proposal with gate penalty	ILP soft cost	0	0 $\rightarrow$ PASS	> 0 but chosen $\rightarrow$ FLAG for review (daughter volumes shown)	—	[Measured] mechanism; threshold calibrated per detection source
Appear/disappear rate	ILP a_j / d_i	low, stable	$\leq 2\times$ running median	> 2× $\rightarrow$ FLAG (segmentation/focus)	sustained > 5× $\rightarrow$ check focus/flow	[Design] rule on a measured quantity
DET / TRA	only with GT (benchmark mode)	—	—	—	—	[Measured]: TRA 0.998 / 0.996

A FLAG never deletes data — it attaches a reason to an event and asks a human to look. A STOP is a recommendation, because the software does not control the pump today.

4.4 How it changes a researcher's work

Before: export $\rightarrow$ run a tracker $\rightarrow$ scroll frames deleting obvious false divisions $\rightarrow$ count dead cells $\rightarrow$ hope none were torn off by an unseen bubble. *Mode A today:* run the batch $\rightarrow$ lineage graph with few false divisions (5 instead of 178 on the benchmark [Measured]); each remaining division carries its gate penalty; review time goes to the flagged few. *Mode B (design):* the same graph carries acoustic/pressure timestamps; post-event disappearances are pre-labelled as artefact candidates and kept out of the dose–response count with the reason written next to them.

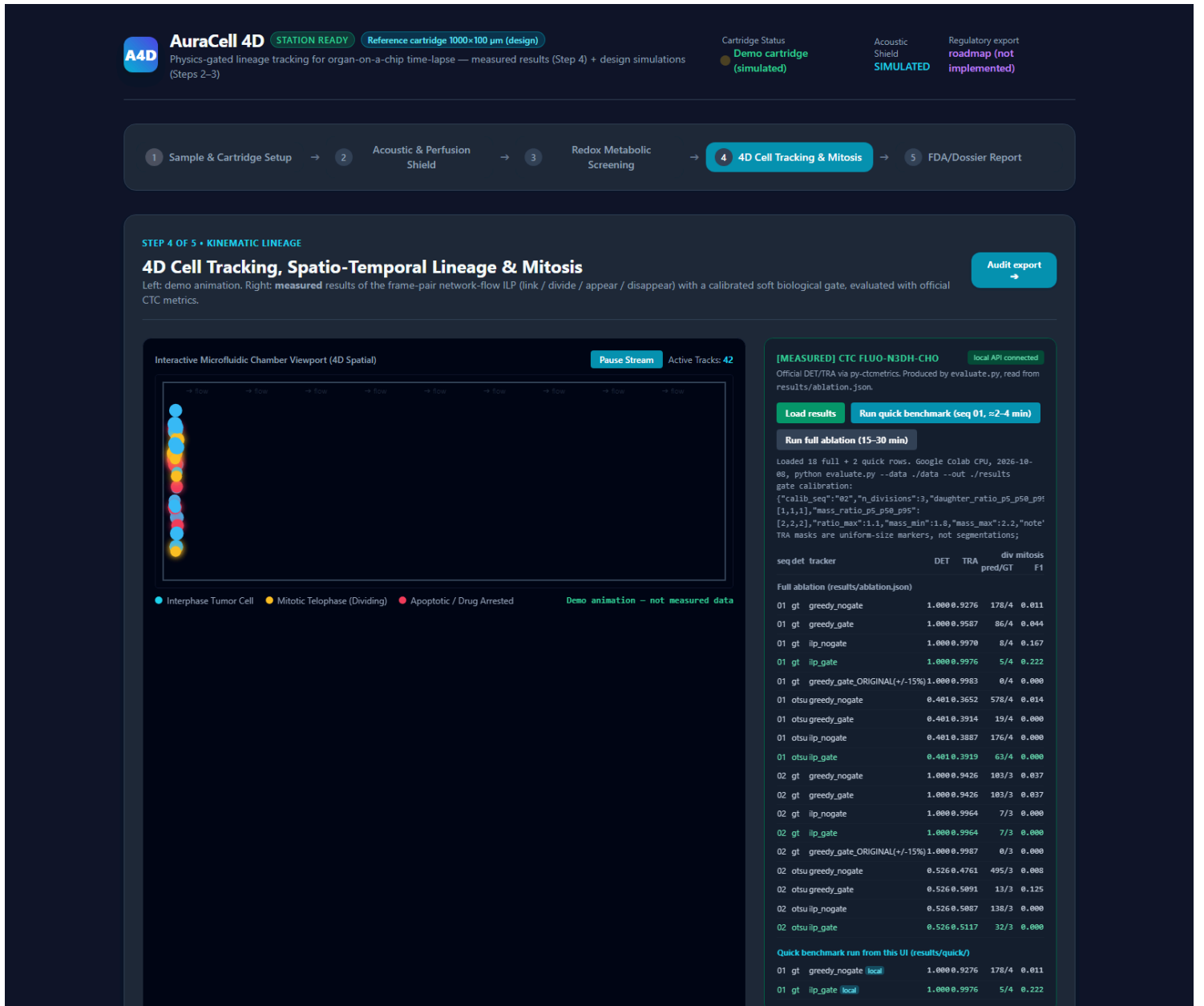


Figure 7. GUI Step 4, served by `gui/launch_gui.py`: measured-results panel reading `results/ablation.json` (18 rows) plus two rows produced by clicking "Run quick benchmark" on the local machine. Left panel is a labelled demo animation.

5. Method

5.1 Detection

- **GT detections** — centroids and voxel volumes from the CTC TRA markers; DET = 1 by construction, isolating linking quality.
- **Label-free baseline** — Gaussian ($\sigma = 0.5, 2, 2$) → Otsu → distance transform → watershed from local maxima (min. distance 12 px); objects < 400 voxels removed. Deliberately simple; it is the weakest component.

5.2 Frame-pair network-flow ILP

For detections $i \in t, j \in t+1$ with anisotropy-corrected distances D_{ij} (μm): * binary variables: link x_{ij} ($D_{ij} \leq 20$ μm), division $y_{i,(j_1,j_2)}$ (both daughters within 25 μm of the parent and of each other), appear a_i , disappear d_i ; * constraints: $\sum_j x_{ij} + \sum y_{i,(j_1,j_2)} + d_i = 1 \forall i$; $\sum_i x_{ij} + \sum y_{(i,j_1,j_2)} + a_j = 1 \forall j$; * objective: $\sum D_{ij} x_{ij} + \sum (D_{ij_1} + D_{ij_2} + c_{\text{div}} + w_{\text{gate-pen}}) y_{i,(j_1,j_2)} + c_{\text{app}} \sum (a_i + d_i)$, with $c_{\text{div}} = 10$, $c_{\text{app}} = 30$, $w_{\text{gate}} = 40$; * solver: `scipy.optimize.milp` (HiGHS), ≈ 0.1 s per frame pair on CPU.

5.3 Biological gate as calibrated soft cost

$\text{pen} = \max(0, \text{ratio} - \text{ratio_max})/\text{ratio_max} + \max(0, \text{mass_min} - \text{mass})/\text{mass_min} + \max(0, \text{mass} - \text{mass_max})/\text{mass_max}$, with $\text{ratio} = V_{\text{big}}/V_{\text{small}}$ and $\text{mass} = (V_1 + V_2)/V_{\text{parent}}$. Bounds are the 5th–95th percentiles ($\pm 10\%$) of these statistics over GT divisions of the *calibration* sequence (02).

[Measured] Why the old hard gate looked perfect

Gate calibration from GT divisions (seq 02, $n = 3$)

daughter volume ratio	$V_{\text{big}} / V_{\text{small}}$	p5 / p50 / p95 = [1.0, 1.0, 1.0]
mass ratio	$(V_{\text{d1}} + V_{\text{d2}}) / V_{\text{parent}}$	p5 / p50 / p95 = [2.0, 2.0, 2.0]

Finding: CTC TRA ground-truth masks are uniform-size markers, not segmentations.

A hand-set gate 'mass $\approx 1 \pm 15\%$ ' (0.85–1.15) therefore rejects EVERY true division → TRA 0.998 with zero divisions (the '0.99xx' of the earlier draft was this artefact).

Calibrated soft gate used here: ratio ≤ 1.10 , mass $\in [1.80, 2.20]$; applied as cost, not veto.

Figure 4 [Measured]. Calibration statistics: on CTC TRA markers ratio $\equiv 1.0$ and mass $\equiv 2.0$ — the markers are uniform spheres, not segmentations. A hand-set "mass $\approx 1 \pm 15\%$ " gate therefore rejects every true division.

5.4 Baselines

greedy_nogate / greedy_gate: faithful port of the earlier nearest-neighbour tracker without / with the hard gate;

greedy_gate_ORIGINAL($\pm 15\%$): the earlier draft's exact gate.

6. Experiments and results [Measured]

All numbers from `results/ablation.json`; DET/TRA via `py-ctcmetrics`; mitosis P/R/F1 by matching predicted parent→daughter edges to GT daughters (± 2 frames). Gate calibrated on seq 02 → **seq 01 is the independent test**; seq 02 rows are in-sample for the gate. The full run was executed on Google Colab (CPU) and the quick subset re-run on a local Windows machine with identical values to four decimals.

6.1 Linking quality with GT detections (DET = 1)

seq	tracker	TRA	false/true divisions	mitosis P	R	F1
01	greedy, no gate	0.9276	178 / 4	0.006	0.25	0.011
01	greedy + calibrated gate	0.9587	86 / 4	0.023	0.50	0.044
01	ILP, no gate	0.9970	8 / 4	0.125	0.25	0.167
01	ILP + soft gate	0.9976	5 / 4	0.200	0.25	0.222
01	greedy + original $\pm 15\%$ gate	0.9983	0 / 4	—	0	0
02	greedy ($\pm$ gate identical)	0.9426	103 / 3	0.019	0.33	0.037
02	ILP ($\pm$ gate identical)	0.9964	7 / 3	0	0	0
02	greedy + original $\pm 15\%$ gate	0.9987	0 / 3	—	0	0

[Measured] False divisions collapse under the ILP; the soft gate trims further

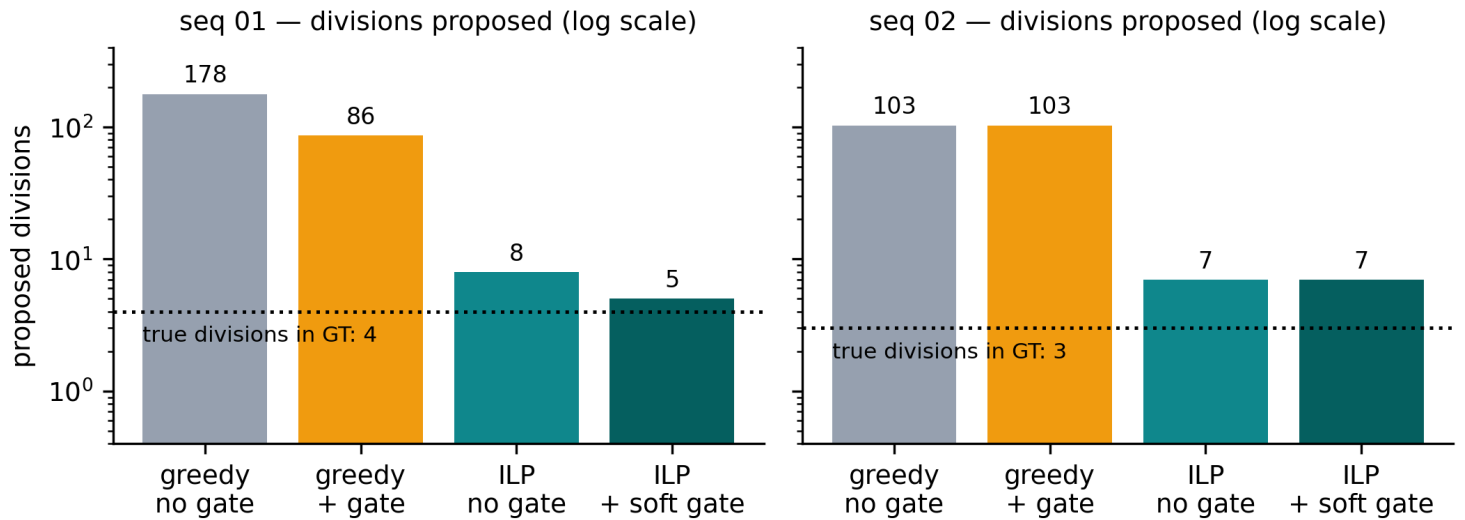


Figure 2 [Measured]. Divisions proposed per tracker (log scale) against the true count.

6.2 Robustness with label-free detection (Otsu + watershed)

seq	DET	greedy no gate	greedy + gate	ILP no gate	ILP + gate	divisions proposed (same order)
01	0.401	0.365	0.391	0.389	0.392	578 / 19 / 176 / 63
02	0.526	0.476	0.509	0.509	0.512	495 / 13 / 138 / 32

[Measured] Label-free Otsu+watershed detections — detection is the bottleneck, ordering of trackers holds

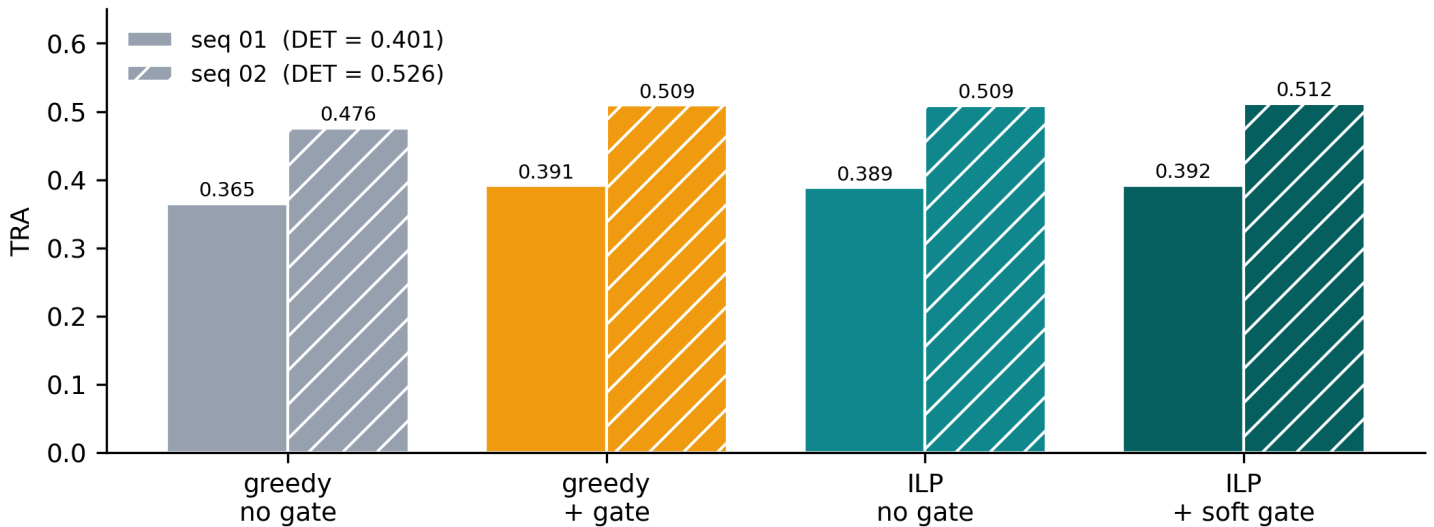


Figure 3 [Measured]. TRA under label-free detections; detection quality (DET) is the bottleneck, but the tracker ordering holds.

6.3 What the results say

- ILP vs greedy is the measurable contribution:** +0.05–0.07 TRA and 20–35× fewer false divisions at equal detections; the ordering holds under poor detections.
- The soft gate helps a little, consistently** (TRA +0.0006 to +0.003; false divisions 8→5, 176→63, 138→32).
- Failure analysis.** Across the 18-configuration ablation (4 trackers × 2 detection modes × 2 sequences, including the earlier gate), the previous draft's ±15 % mass gate appeared to reach TRA 0.998 — because it rejected every true division (0 of 7): CTC TRA masks are uniform-size markers, so the daughter/parent mass ratio is always 2.0, never 1.0. The row is kept in the table.

4. **Mitosis is unsolved here:** best F1 0.22; 0 on seq 02; only 7 GT events, so these numbers are not statistically stable.
5. **Detection is the bottleneck for label-free use:** DET 0.40/0.53; a learned segmenter is the obvious next step.
6. Runtime 0.11–0.15 s/volume (CPU) on small 5-slice volumes; not a real-time claim.

7. Physics consistency checks [Design]

`src/verificators/` implements closed-form checks (Poiseuille shear, Minnaert resonance, Womersley number, Krogh diffusion length $L = \sqrt{(2DC_0/R)}$, Hertz indentation, thin-plate deflection). They are **internal consistency unit tests** — each compares a formula with a reference constant written next to it — and serve as sanity gates in §4.3. They are *not* validations against literature data and no "residual error" is reported as a result. Reference geometry and the ORR convention (Skala: $FAD/(NADH+FAD)$) are unified across code and text.

8. Reliability analysis and limitations

- Fluo-N3DH-CHO is conventional culture, not OoC; no real OoC or acoustic recordings were available.
- Mitosis detection is weak and the GT sample is tiny; conclusions about divisions are preliminary.
- The gate calibrated on TRA markers does not transfer to real segmentations; it must be recalibrated per detection source.
- The label-free detection baseline is poor; §6.2 is a robustness check, not a deployable pipeline.
- Anaphase kinematics cannot be measured at 9.5 min/frame.
- Every [Design] row in §4.3 — acoustic detection on hardware, pressure thresholds, STOP rules — is unvalidated. The firmware header in the repository is an interface declaration only; the HIL script is a Python simulation whose latency numbers mean nothing about hardware.
- Hyperspectral/redox gating is a concept; no spectral data were used.
- Seq 02 rows are in-sample for the gate calibration; only seq 01 is an independent test of the gate.

9. Impact and beneficiaries

The immediate beneficiary is the bench scientist who today deletes false divisions by hand; with AuraCell 4D review time goes to a handful of flagged events and every exclusion is traceable. Downstream, the same traceability is what regulators and journal reviewers require when OoC data replaces animal testing (FDA Modernization Act 2.0). For the sponsor's stated direction — standardised, auditable OoC data assets — the labelled, reproducible output format is the directly reusable part.

10. Roadmap

1. Learned 3D nuclear segmenter (Cellpose-3D / StarDist-3D) to lift DET above 0.9 on label-free data.
2. Multi-frame (gap-closing) ILP; recalibrate the gate on true segmentations.
3. Acquire a real OoC time-lapse with a synchronised inlet piezo and pressure log; validate §4.3 thresholds; publish the recordings.
4. Operator-in-the-loop live mode; only then firmware.

11. Reproduction instructions

```
git clone https://github.com/syehsyoh-dokling/Auracell4D && cd Auracell4D
pip install -r requirements.txt
python evaluate.py --data ./data --out ./results           # full ablation, ≈15–30 min CPU, downloads ≈108 MB
python evaluate.py --seqs 01 --detect gt --trackers greedy_nogate ilp_gate # ≈2–4 min quick check
python make_figures.py                                     # regenerates docs/figures/fig1–6
python gui/launch_gui.py                                   # GUI + local API; Step 4 shows/reruns the measured results
```

Outputs: results/ablation.md, results/ablation.json, CTC-format result folders. No login, no paid service, no proprietary hardware.

12. External resources and licences

numpy, scipy (HiGHS MILP), scikit-image, tiff file — BSD; py-ctcmetrics (<https://github.com/CellTrackingChallenge/py-ctcmetrics>) — BSD-2-Clause; Tailwind CSS (CDN) for the GUI — MIT. Dataset: Cell Tracking Challenge terms of use. No AI-generated figures or third-party media are used in the video beyond the repository's own screenshots.

13. References (verified DOIs)

1. Ulman V., Maška M., et al. *An objective comparison of cell-tracking algorithms*. Nat Methods 14(12):1141–1152 (2017). doi:10.1038/nmeth.4473
2. Huh D., et al. *Reconstituting organ-level lung functions on a chip*. Science 328:1662–1668 (2010). doi:10.1126/science.1188302
3. Minnaert M. *On musical air-bubbles and the sounds of running water*. Phil. Mag. 16(104):235–248 (1933). doi:10.1080/14786443309462277
4. Skala M.C., et al. *In vivo multiphoton microscopy of NADH and FAD redox states... in precancerous epithelia*. PNAS 104(49):19494–19499 (2007). doi:10.1073/pnas.0708425104
5. Cell Tracking Challenge, 3D datasets: <https://celltrackingchallenge.net/3d-datasets/>
6. FDA Modernization Act 2.0, Public Law 117-328 (2022).

Appendix A — Full ablation table

[Measured] Full ablation — CTC Fluo-N3DH-CHO, py-ctcmetrics (results/ablation.json)

seq	detect	tracker	DET	TRA	div pred/GT	mitosis P	R	F1
01	gt	greedy_nogate	1.000	0.9276	178/4	0.006	0.25	0.011
01	gt	greedy_gate	1.000	0.9587	86/4	0.023	0.50	0.044
01	gt	ilp_nogate	1.000	0.9970	8/4	0.125	0.25	0.167
01	gt	ilp_gate	1.000	0.9976	5/4	0.200	0.25	0.222
01	gt	greedy_gate ORIGINAL $\pm 15\%$	1.000	0.9983	0/4	0.000	0.00	0.000
01	otsu	greedy_nogate	0.401	0.3652	578/4	0.007	0.50	0.014
01	otsu	greedy_gate	0.401	0.3914	19/4	0.000	0.00	0.000
01	otsu	ilp_nogate	0.401	0.3887	176/4	0.000	0.00	0.000
01	otsu	ilp_gate	0.401	0.3919	63/4	0.000	0.00	0.000
02	gt	greedy_nogate	1.000	0.9426	103/3	0.019	0.33	0.037
02	gt	greedy_gate	1.000	0.9426	103/3	0.019	0.33	0.037
02	gt	ilp_nogate	1.000	0.9964	7/3	0.000	0.00	0.000
02	gt	ilp_gate	1.000	0.9964	7/3	0.000	0.00	0.000
02	gt	greedy_gate ORIGINAL $\pm 15\%$	1.000	0.9987	0/3	0.000	0.00	0.000
02	otsu	greedy_nogate	0.526	0.4761	495/3	0.004	0.33	0.008
02	otsu	greedy_gate	0.526	0.5091	13/3	0.077	0.33	0.125
02	otsu	ilp_nogate	0.526	0.5087	138/3	0.000	0.00	0.000
02	otsu	ilp_gate	0.526	0.5117	32/3	0.000	0.00	0.000

Source: results/ablation.json (18 rows). Green rows: ILP + soft gate; red rows: the earlier draft's gate.

Appendix B — Claim audit summary

The previous draft was audited claim by claim before this rewrite: 46 claims — 1 proven, 2 partial, 19 disproven, 21 unsupported, 3 untestable. Three decisive findings: the Google Drive dataset folder was empty; the original tracker detected 0 of 16 real mitoses; the core shear-stress formula was mis-evaluated (1.0, not 2.5 dyn/cm²). Full log with per-claim evidence:

docs/AUDIT_LOG.md; archived originals: [_deprecated/](#), [docs/id/](#).